



ACADEMIC EXPANSION STRATEGY: DRIVING UNIVERSITY ADOPTION ACROSS GERMANY

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SDI University 2026
Customer Centricity Module

AGENDA

Introduction

Target Cross

Market segmentation

Persona models

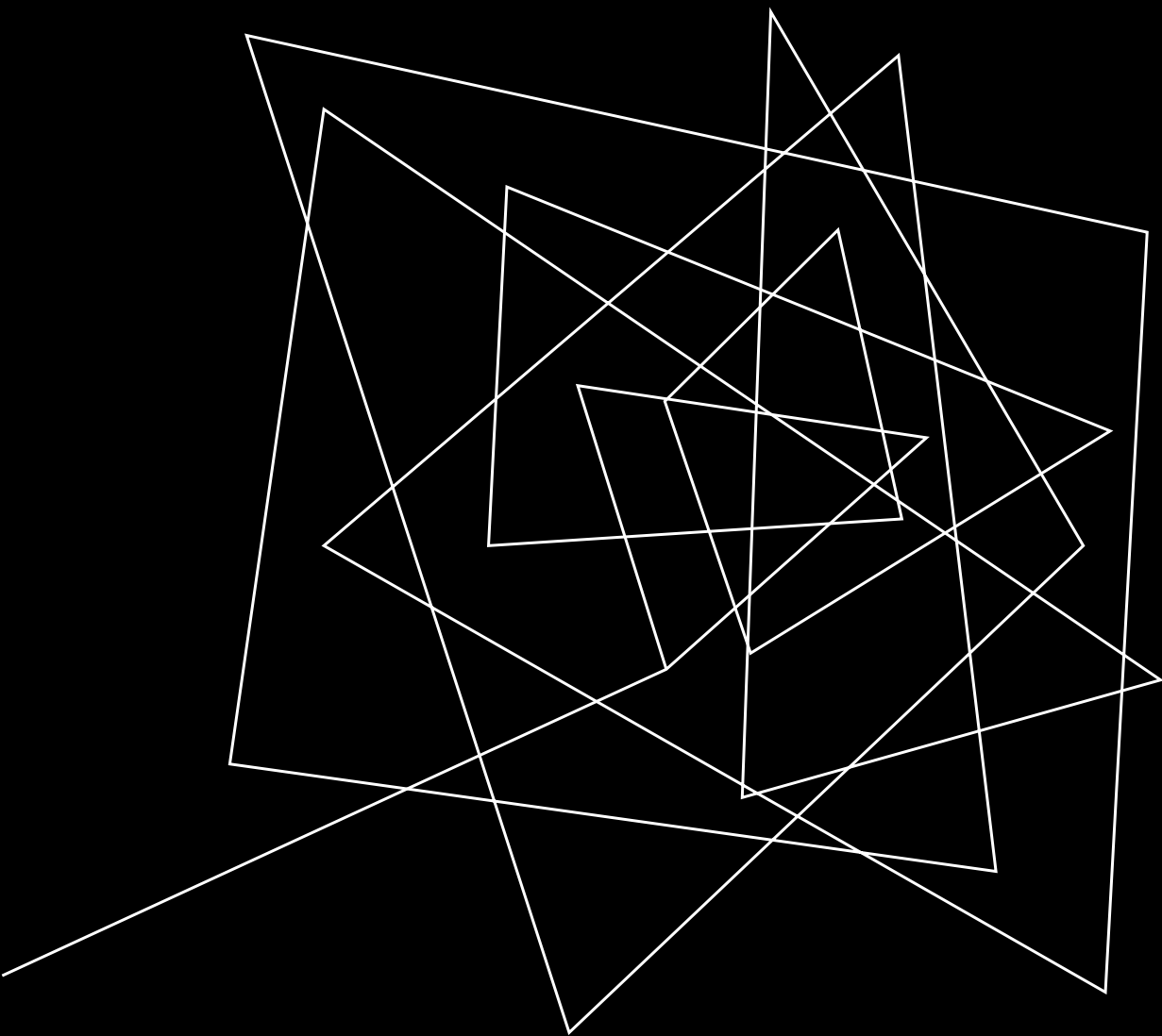
Value Proposition

Exam Question – Use Case

Customer Journey Map

Keywords

Conclusion



TARGET-CROSS TABLE

TARGET CROSS

	WHY/ Value		WHO/ Users	
	• Expand Hanser's digital academic portfolio with an AI-based tool • Support universities in digitalization & modern teaching • Provide a citable, reliable alternative to general AI tools • Capture long-term institutional revenue		• Professors (lecture creation & supervision) • Research Group Leaders (methods, literature, lab-support) • IT Administrators • Master Students (advanced academic users) • Bachelor Students (exam & lab report support) • Institutions = paying customers	
	WHAT/ Offering		KPIs/ Success metrics	
	• Academic market analysis & segmentation • Persona-based value propositions • Use-case comparison Plastics.ai vs ChatGPT • Pricing models (Campus License, Seat Packages) • Customer Journey design • Keyword analysis for awareness • Integration with eCampus & Dozierendenportal		• Universities onboarded • ARR from academic segment • Professor & student active usage • Courses/labs adopting Plastics.ai • Number of theses citing Plastics.ai • Onboarding time & system stability	

IDENTIFY AND SEGMENT THE TARGET UNIVERSITIES

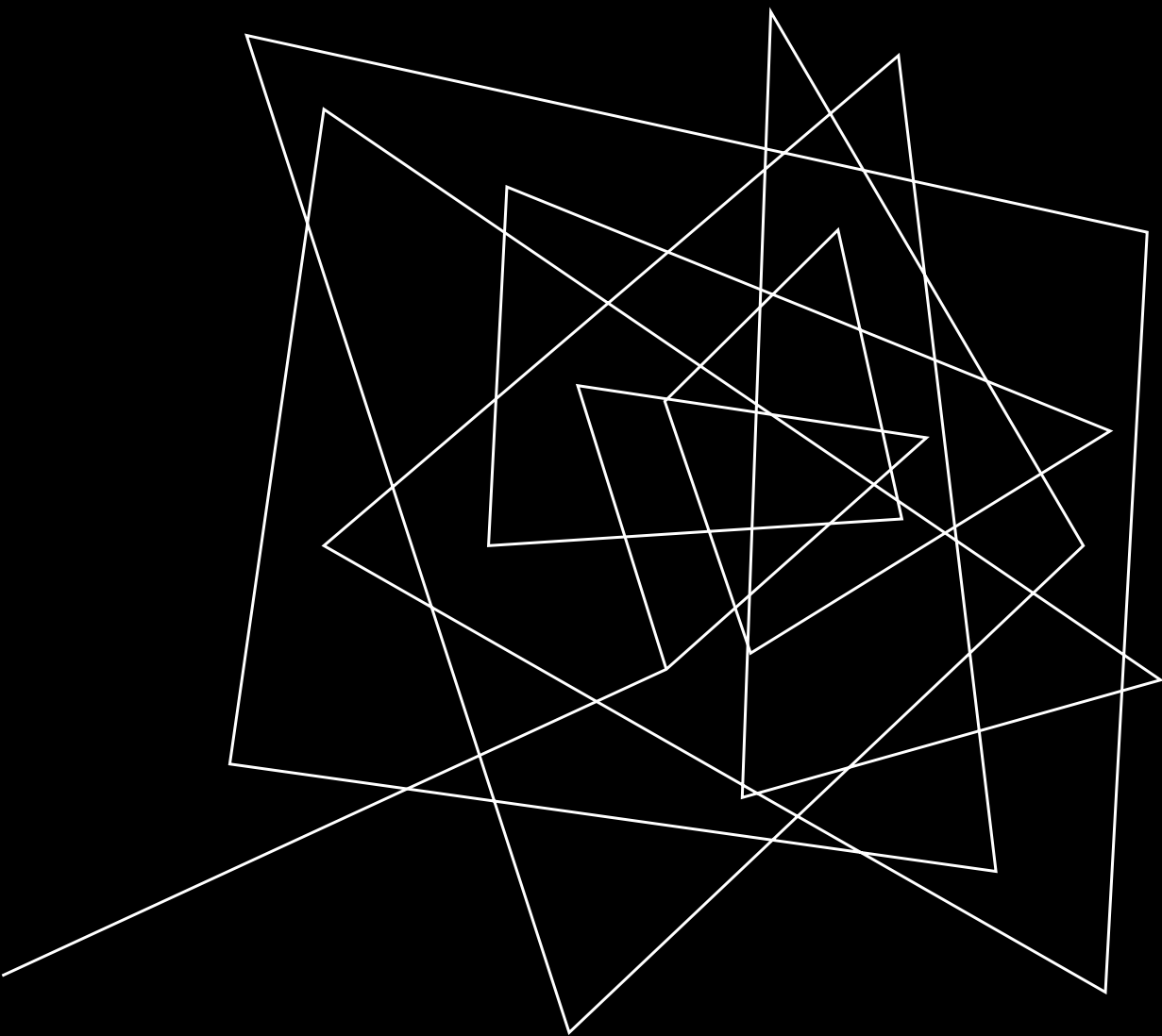
Universities (Technische Universitäten - TU / Universitäten - Uni):

- **RWTH Aachen University** (Leading in engineering)
- **TU Munich (TUM)**
- **University of Bayreuth** (World-renowned for Polymer Science)
- **TU Darmstadt**
- **TU Dresden**
- **University of Stuttgart**
- **Friedrich-Alexander-Universität Erlangen-Nürnberg (FAU)**

Universities of Applied Sciences (Hochschulen für Angewandte Wissenschaften - HAW):

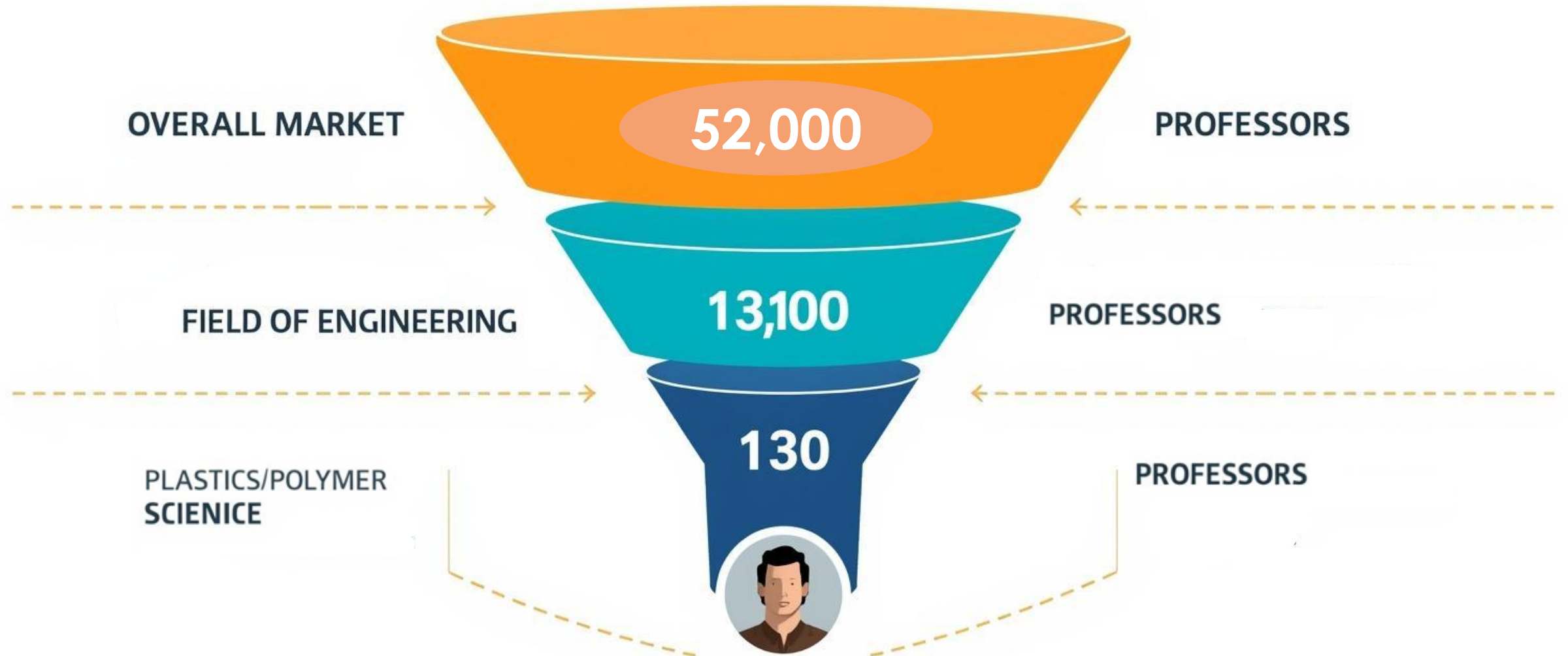
- **Munich University of Applied Sciences (HM)**
- **Aachen University of Applied Sciences (FH Aachen)**
- **Cologne University of Applied Sciences (TH Köln)**
- **Hochschule Merseburg** (Strong in Chemistry and Plastics)
- **Hochschule Niederrhein** (Strong in Textile and Polymer Engineering)

	Number	Source
1. Total Engineering students	~748,000	Too few engineering students in Germany: Mechanical engineering particularly affected
2. Higher-education institutions	422 ($\approx$ 109 universities; $\approx$ 209 UAS)	Hochschulen in Deutschland CHE Hochschuldaten
3. Professors (field of engineering)	~13,240 (130 specialized in plastics/polymer science)	White Paper: Polymer Research in Germany
4. Study locations offering plastics/polymer programs	~25–26	Wo kann man Kunststofftechnik studieren? Alle Städte und Unis
5. Concrete programs	~9–26 B.Sc. and ~9–15 M.Sc. programs	Bachelor Kunststofftechnik 2025/2026: Alle Studiengänge
6. Estimated students in dedicated plastics programs	~2,000 – 4,000 total enrolled	Calculations based on given numbers



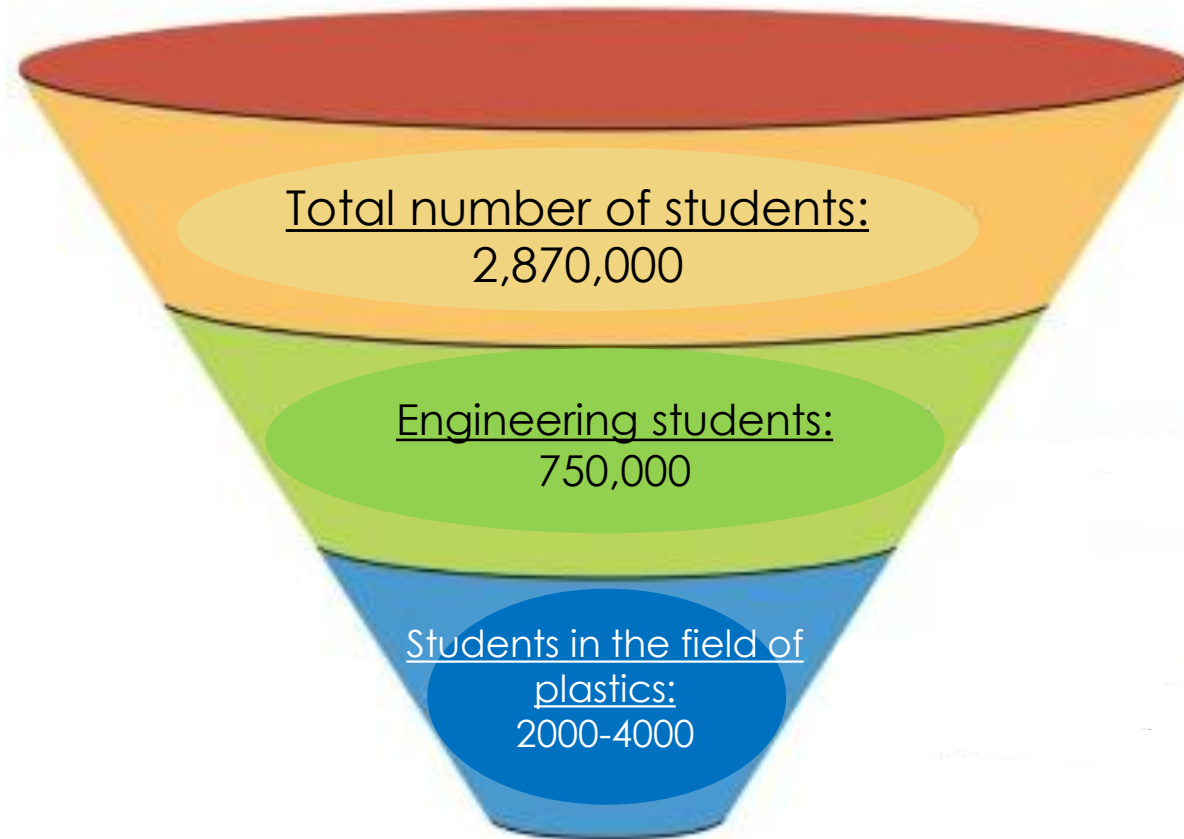
ACADEMIC PERSONA MODEL

Market Funnel: Professors
(Germany, 2024)



Persona: **Prof, Dr. Markus Weber**

Market Funnel: Students (Germany, 2024)



Persona: **Julia Hartmann**

Persona 1 – Professor (Technical University)

Name: Prof. Dr. Markus Weber

Age: 52

Gender: Male

Location: Munich, Germany

Profession: Chair of Plastics Engineering, TUM

Tasks:

- Supervises Master's & PhD students
- Leads research projects
- Prepares lectures & labs

Problems:

- Literature research takes too long
- Students use unreliable AI tools
- Pressure to publish quickly

Solution with plastics.ai:

- Citable expert answers
- Faster research & teaching prep
- Consistent knowledge base

Wishes: Digital tools that reduce workload

Values: Accuracy, reliability, efficiency

CVP: "plastics.ai accelerates research & teaching with trustworthy expert knowledge."

Budget: €3,000–10,000

Media Usage: LinkedIn, ResearchGate, journals

Jobs to be done:

1. Design and update lecture content in plastic engineering
2. Writing scientific papers
3. Supervising bachelor, master and Phd thesis

Solution of plastics.ai:

1. Offers fast, correct and citable teaching content
2. Citable background and theoretical frameworks
3. Structured literature insights and reserach guidance

Persona 2 – Junior Research Group Leader

Name: Dr.-Ing. Anna Richter

Age: 38

Gender: Female

Location: Aachen, Germany

Profession: Junior Professor, RWTH Aachen

Tasks:

- Supervises students
- Writes grant proposals
- Organizes lab research

Problems:

- High supervision load
- Students struggle with literature
- Need faster onboarding

Solution with plastics.ai:

- Cuts supervision time
- Provides quick, reliable explanations

Wishes: More time for publications

Values: Innovation, scientific quality

CVP: "plastics.ai reduces supervision time and drives research excellence."

Budget: €1,500–6,000

Media Usage: Google Scholar, LinkedIn

Jobs to be done:

1. Design and coordinate own research project
2. Run and manage lab resources, safety and compliance

Solution of plastics.ai:

1. Provides an overview of methods/checklists based on textbooks
2. Provides guidance of instrumentation and safety documentation

Persona 3 – IT Administrator (University of Applied Sciences)

Name: Thomas Klein

Age: 47

Gender: Male

Location: Aachen, Germany

Profession: IT Manager, Aachen UAS

Tasks:

- Selects digital tools
- Ensures data security
- Manages licenses

Problems:

- GDPR risks with AI tools
- Pressure to modernize
- Need affordable tools

Solution with plastics.ai:

- GDPR-compliant
- Simple licensing
- Easy integration

Wishes: Stable, secure solutions

Values: Security, cost efficiency

CVP: "plastics.ai brings secure, compliant AI to higher education."

Budget: €5,000–20,000

Media Usage: IT magazines, LinkedIn

Jobs to be done:

1. Ensure reliable IT services and infrastructure
2. Data protection and legal compliance

Solution of plastics.ai:

1. Private cloud hosted in Germany
2. Provide data processing agreement and privacy impact assessment

Persona 4 – Master Student (Technical University)

Name: Julia Hartmann

Age: 24

Gender: Female

Location: Stuttgart, Germany

Profession: Master Student, Plastics Engineering

Tasks:

- Writes lab reports
- Thesis research
- Learns complex polymer concepts

Problems:

- Hard to find reliable information
- Coursework overload
- AI tools give incorrect info

Solution with plastics.ai:

- Clear, citable explanations
- Faster learning
- Better lab preparation

Wishes: Reliable explanations

Values: Accuracy, clarity

CVP: "plastics.ai helps students master complex topics effortlessly."

Budget: €0–50

Media Usage: YouTube, Instagram, LinkedIn

Jobs to be done:

1. Conduct fast literature review; find reliable information/sources
2. Design experiments; write lab protocols

Solution of plastics.ai:

1. Offers short summaries and citation lists (curated literature overview)
2. Instant protocol templates and checklists with references to standard test methods and textbooks

Persona 5 – Bachelor Student (Applied Sciences)

Name: Daniel Meyer

Age: 21

Gender: Male

Location: Rosenheim, Germany

Profession: Bachelor Student, Applied Plastics Technology

Tasks:

- Learns polymer processing basics
- Prepares for labs & exams
- Works on applied projects

Problems:

- Theory is difficult
- Needs simple explanations
- Limited literature access

Solution with plastics.ai:

- Simplified but accurate answers
- Connects theory to practice
- Supports labs & projects

Wishes: Accessible knowledge

Values: Practical relevance, clarity

CVP: "plastics.ai makes engineering knowledge understandable & applicable."

Budget: €0–50

Media Usage: YouTube, TikTok, ILIAS

Jobs to be done:

1. Conduct fast literature review; find reliable information/sources
2. Design experiments; write lab protocols

Solution of plastics.ai:

1. Offers short summaries and citation lists (curated literature overview)
2. Instant protocol templates and checklists with references to standard test methods and textbooks

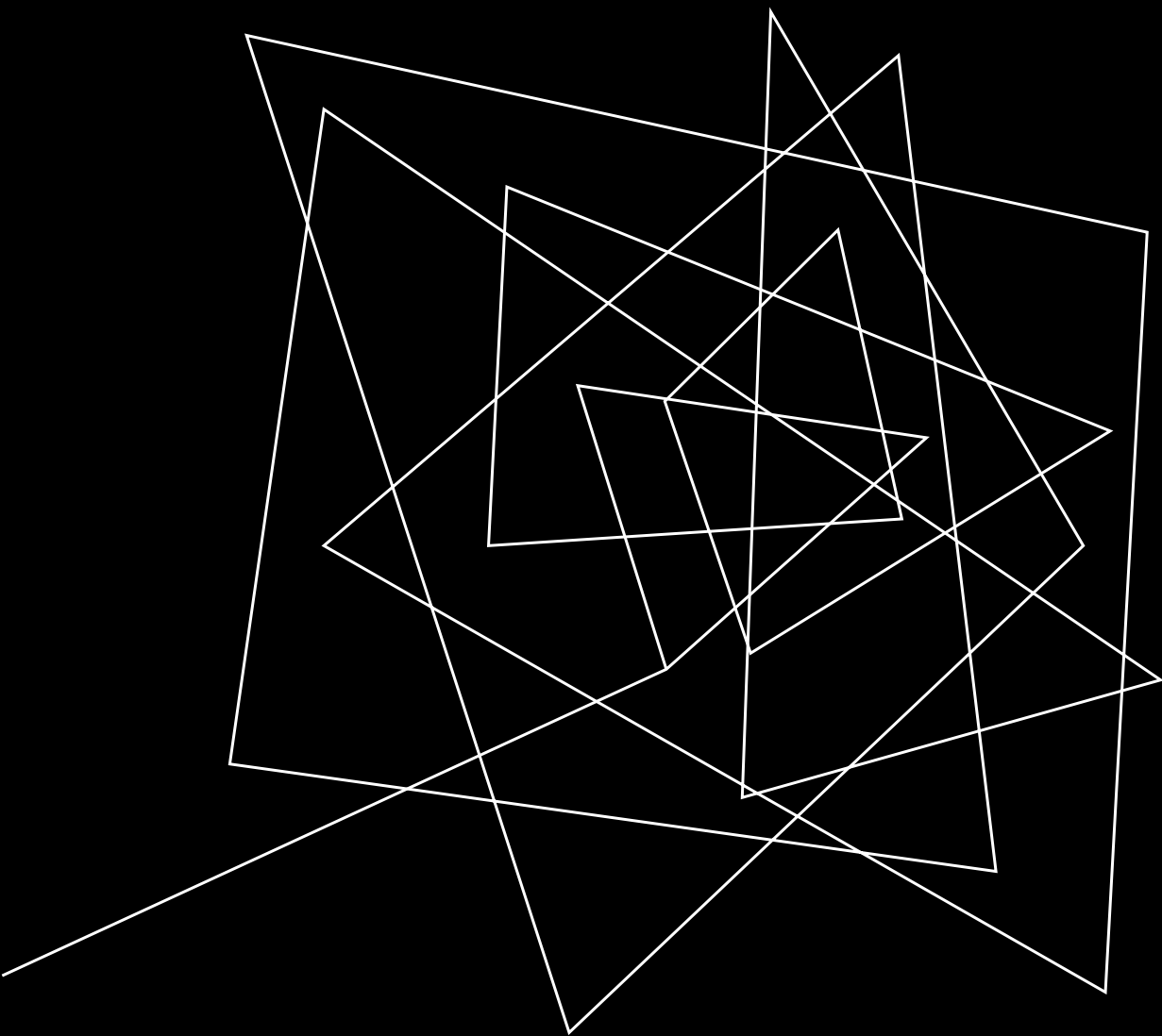
Reasons why the Personas would pay for plastics.ai:

	<u>Why they would pay:</u>	<u>Key tasks:</u>	<u>Plastics.ai – solution:</u>
 Professor:	-reduce teaching workload -responsibility for large student groups	-prepare & update lectures -supervise theses with limited time	-offers validated definitions & explanations -access to citable and textbook-based sources
 Students:	-pressure to perform in exams & thesis writing	-literature research & understand complex concepts	-reliable, citable sources for academic writing & faster literature orientation



Most valuable target group: **Universities as Institutions**

- Scalability: one campus license reaches hundreds of students
- Predictable and recurring revenue: universities work with annual budgets and multi-year contracts (→high planning security)
- Strategic embedding: integration into library services and digital teaching strategies



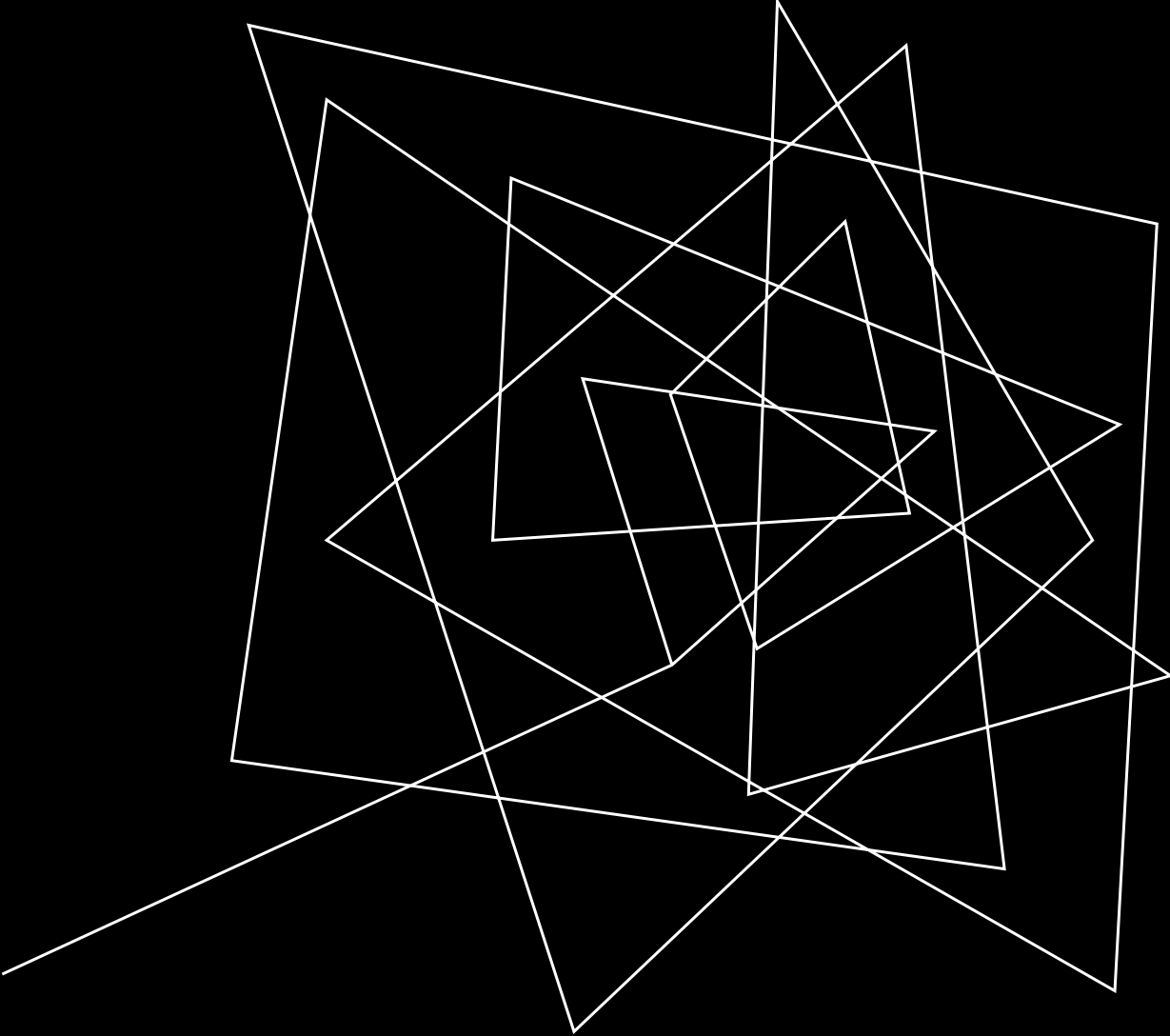
VALUE PROPOSITION

Persona	Role in Ecosystem	Core Value Proposition	Indicative Scale (DE)
Professors	Academic gatekeepers	Teaching & supervision support with textbook-aligned, citable explanations	~130 professors (plastics / polymer science)
Junior Research Group Leaders	Supervision & research quality	Reduced supervision effort through consistent academic explanations	~520 junior researchers (plastics-related)
Bachelor Students	High-volume learners	Exam-focused, exam-safe answers using expected language	~2,000–4,000 students (plastics programs)
Master Students	Advanced academic users	Reliable foundations for coursework and theses	~2,000–4,000 students (plastics programs)
IT Administrators	Institutional enablers	Controlled, compliant AI rollout at university level	~25–35 institutions

A Natural Extension of Hanser's Digital Ecosystem

- Natural fit with the HANSER ecosystem
- Complements the HANSER Dozierendenportal
- Complements the HANSER eCampus
- Lower adoption barriers





EXAM QUESTION – USE CASE: PLASTICS.AI VS CHATGPT

Exam Question: Causes and Corrections for Severe Warpage in PP Injection Molding

Prompt to plastics.ai

answer this one: A polypropylene (PP) part shows severe warpage after injection molding. Explain the three most likely causes of warpage and describe how each cause can be corrected by adjusting machine parameters or mold design. Provide specific parameter adjustments and describe the industrial reasoning behind them.

Answer from plastics.ai

Causes and Corrections for Severe Warpage in PP Injection Molded Parts

1. Temperature Gradients in the Mold

****Cause:****

Nonuniform mold temperature—such as differences between mold halves, cores, or across the cavity—leads to uneven cooling rates. This creates temperature gradients through the wall thickness or across the part, resulting in nonuniform shrinkage and internal stresses that cause warpage. [1][3]

****Correction:****

- ****Balance Mold Temperature:****
- Adjust and test different temperatures on mold halves and cores to achieve uniform cooling.
- Use additional cooling channels or a cooling fixture if necessary.
- Industrial reasoning: Uniform cooling ensures even solidification and shrinkage, minimizing internal stresses and warpage. [2]

Prompt to ChatGPT:

answer this one: A polypropylene (PP) part shows severe warpage after injection molding. Explain the three most likely causes of warpage and describe how each cause can be corrected by adjusting machine parameters or mold design. Provide specific parameter adjustments and describe the industrial reasoning behind them.

Answer from ChatGPT:

Causes and Corrections for Severe Warpage in PP Injection Molded Parts

1) Differential shrinkage from unbalanced packing (gate-side overpack / far-side underpack)

What happens (why it warps)

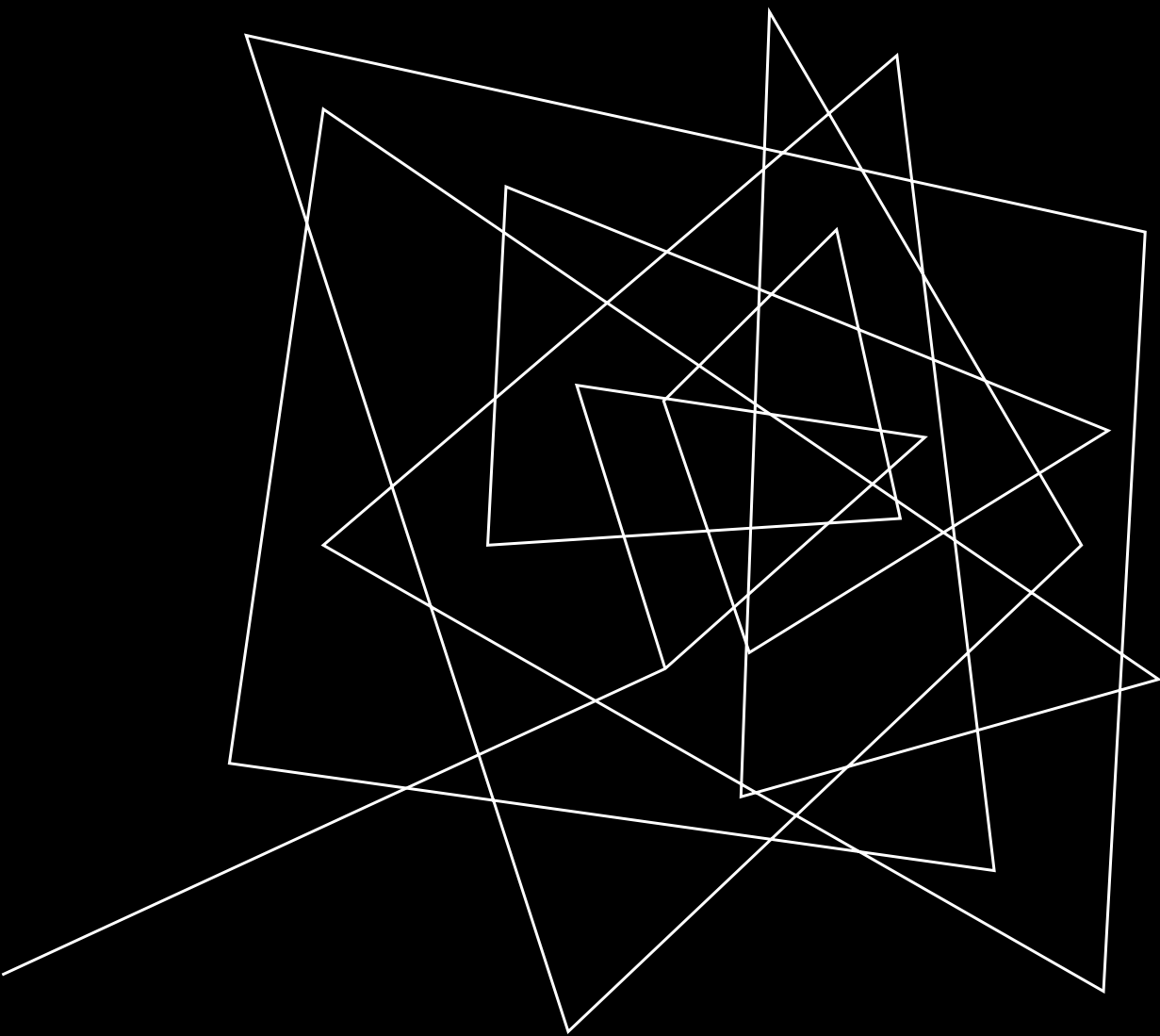
PP shrinks a lot as it crystallizes. If one region is denser (more packed) than another, the part shrinks unevenly after ejection → it bends toward the side with greater shrink (often the underpacked side), or twists if the imbalance is across the part.

Machine parameter corrections (specific)

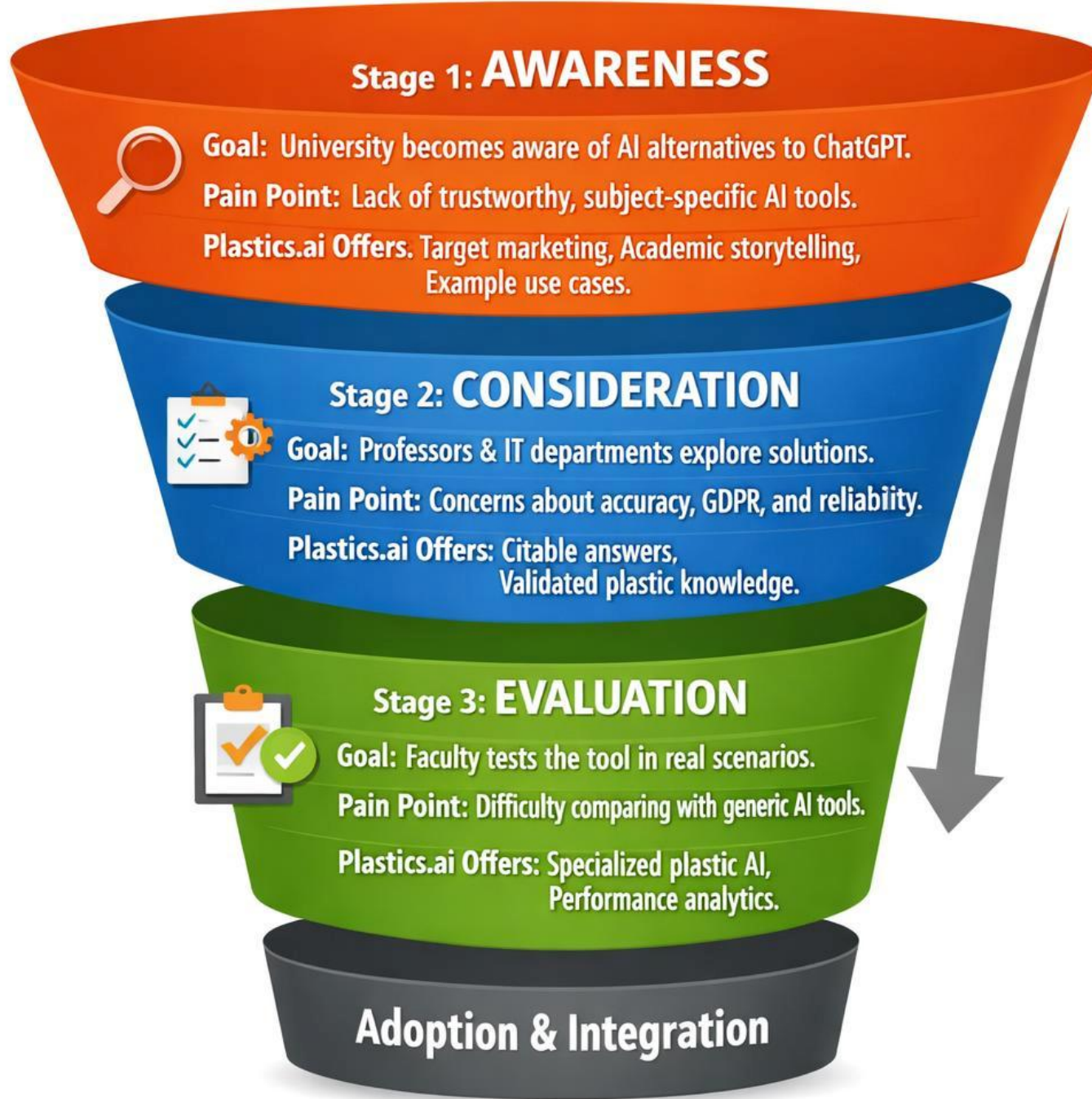
- Set V→P transfer earlier (closer to ~95–98% filled by position, not by time).
Reasoning: prevents “filling in pack,” which over-compresses the gate region and freezes stress.
- Reduce hold pressure in steps (e.g., drop 10–20% at a time) while watching sinks/weight.
Reasoning: minimizes overpacking near the gate (a classic warpage driver in PP).

Why plastics.ai Is the Better Solution for a Common Exam Question

Dimension	plastics.ai Solution (Better for Exams)	ChatGPT solution
Answer Structure	Clean 3-cause → correction structure	Multi-layered, detailed, process-driven
Alignment with Exam Expectations	Very high – mirrors textbook & lecture notes	Moderate – goes beyond what exams usually require
Clarity of Root Causes	Causes are clearly separated and labeled	Causes are technically correct but more intertwined
Language Style	Textbook / academic wording	Industrial, practitioner-oriented language
Parameter Adjustments	Simple, direct (increase/decrease, balance)	Detailed, stepwise, nuanced
Cognitive Load for Students	Low – easy to memorize and reproduce	Higher – requires understanding and synthesis
Risk of Over-Answering	Low – concise and focused	High – may exceed time/space limits
Consistency with Standard Plastics Texts	Very high (cooling, packing, geometry)	High but framed differently
Cause–Effect–Correction Logic	Linear and explicit	More realistic but less linear
Industrial Reasoning	Short, generalized reasoning	Deep, mechanism-based reasoning

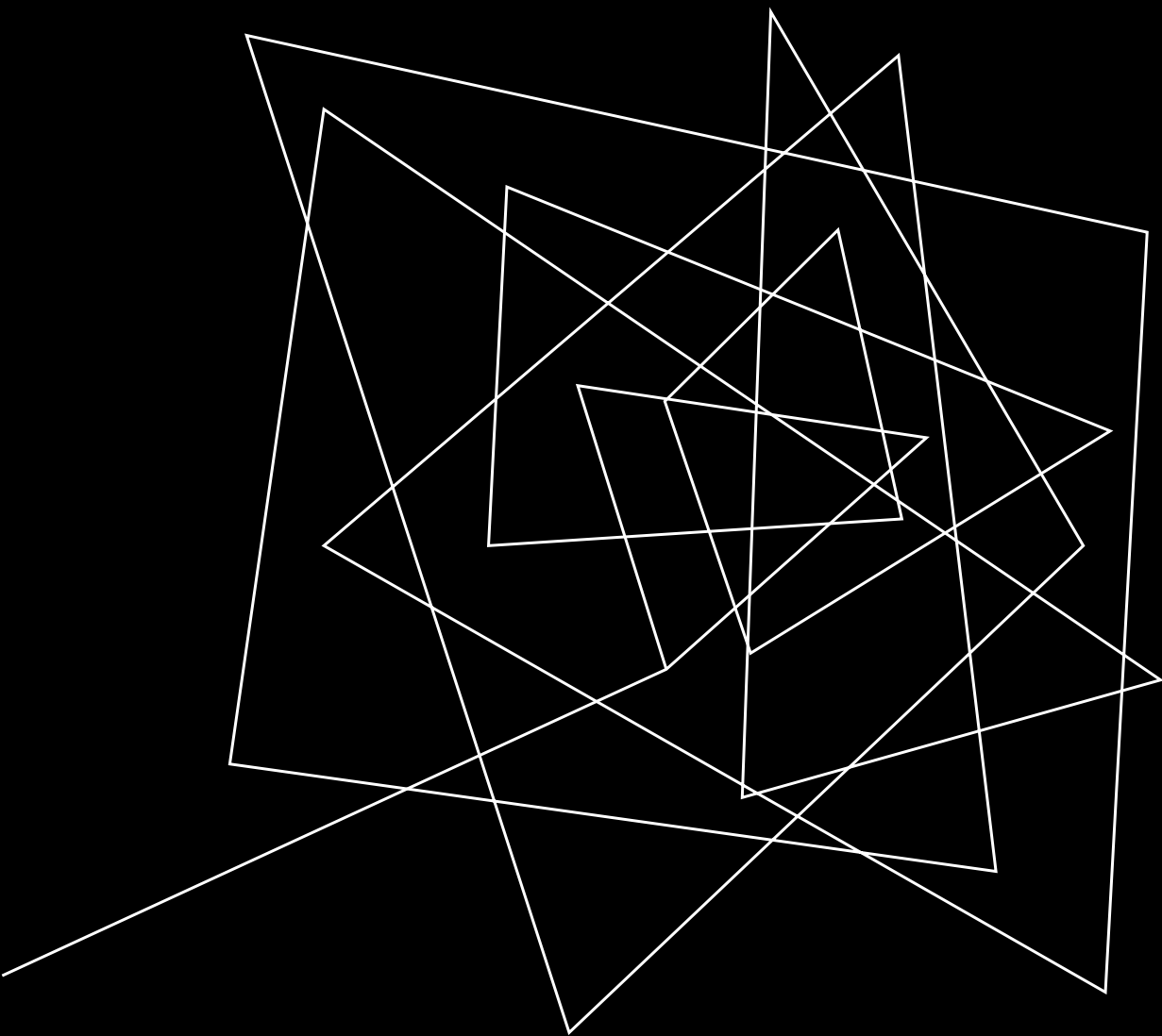


CUSTOMER JOURNEY MAP



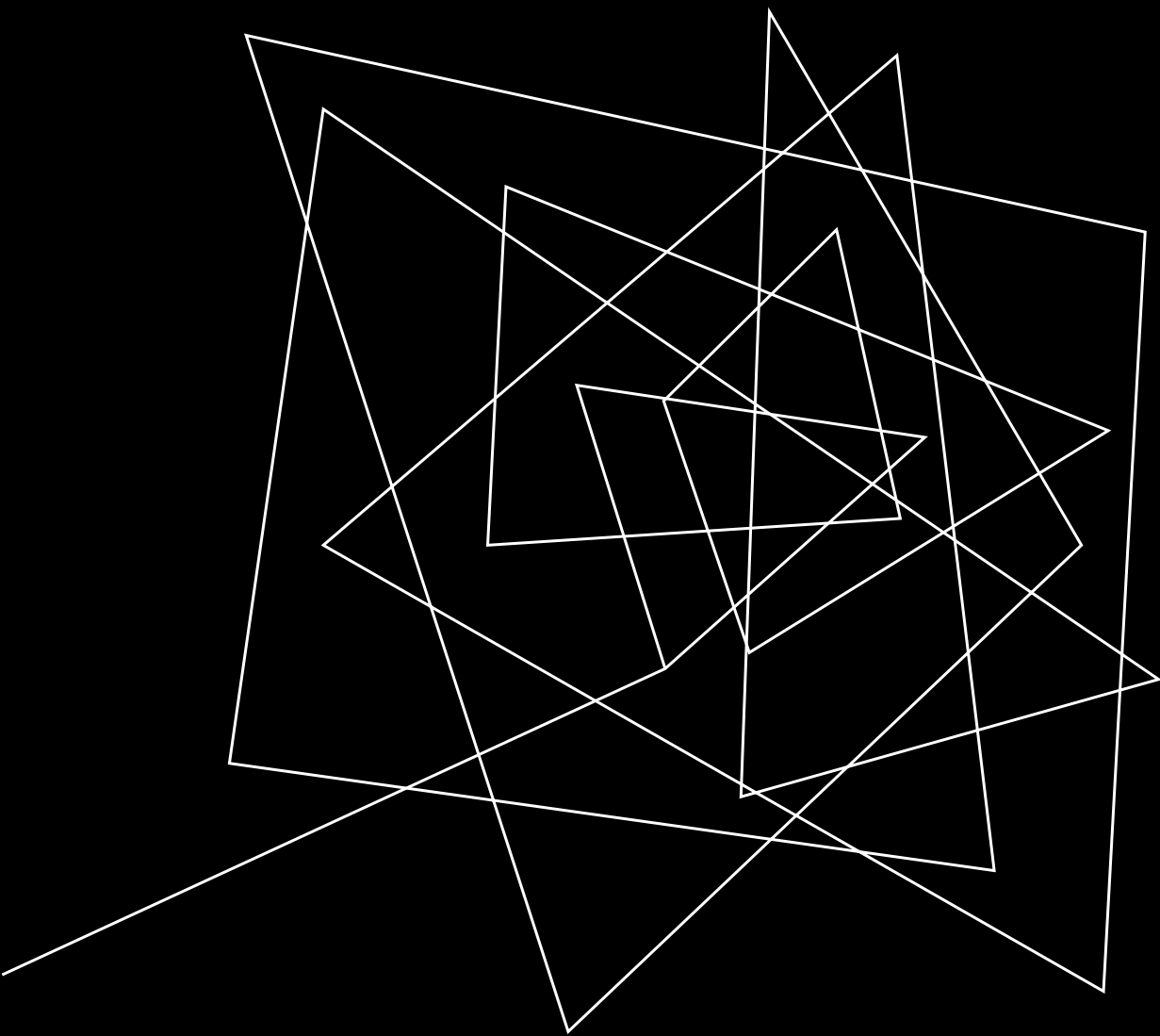
Plastic-AI Academic Validation

Stage	Goal	Pain Point	What Plastics.ai offers
1. Awareness	University becomes aware of AI alternatives to ChatGPT	Lack of trustworthy, subject-specific AI tools	Target marketing, academic storytelling, example use cases
2. Consideration	Professors & IT explore solutions	Concerns about accuracy, GDDR, reliability	Citable answers, validated plastic knowledge
3. Evaluation	Faculty tests tool in real scenarios	Difficulty comparing with generic AI tools	Use-case comparison
4. Pilot	Small group of students/professors try it	Need measurable impact	Free/low-cost semester license, onboarding support
5. Adoption	University purchases campus or seat licenses	Budget approval, technical integration	Simple deployment, predictable pricing, integration into teaching
6. Advocacy	Professors and students recommend the tool	Need proof of long-term value	Citations in thesis, workshops, case studies from partner universities



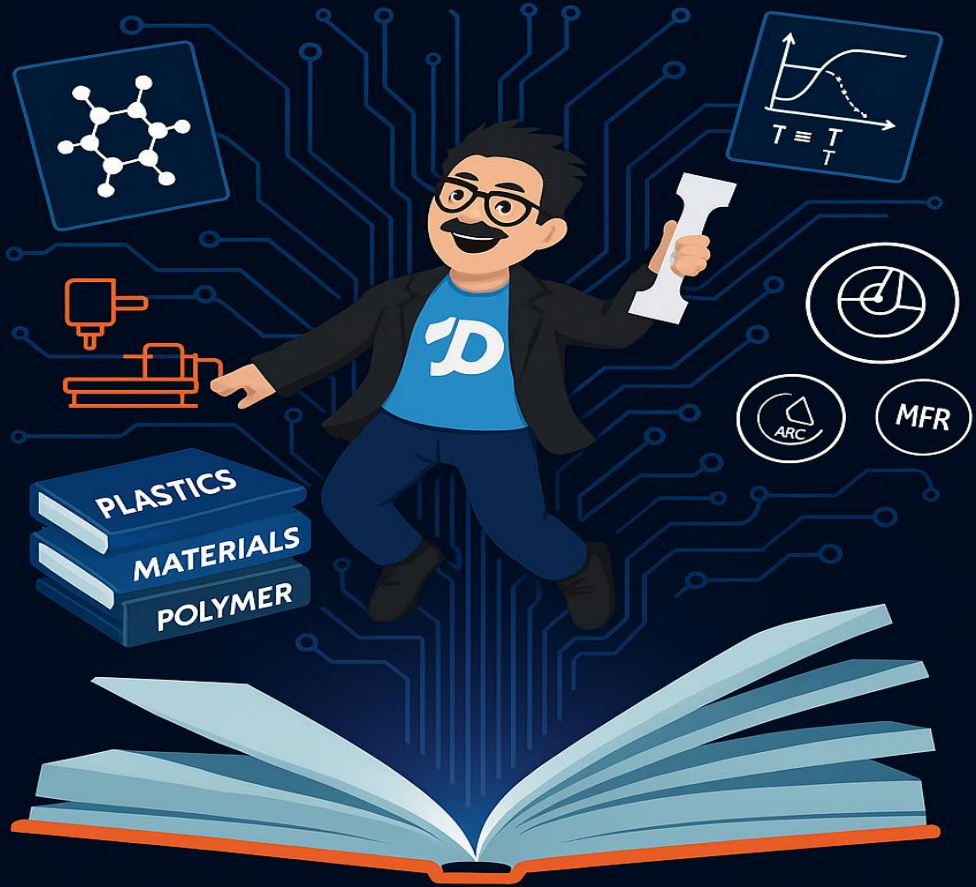
KEYWORDS

Purpose	Keyword Cluster	Core Keywords	Source of Keywords
Academic fields	Core disciplines	Polymer Science • Plastics Engineering • Materials Science	University department & program titles
Courses & exams	Fundamentals	Polymer structure & properties • Thermoplastics vs thermosets	Course syllabi & exam outlines
Processing topics	Manufacturing	Injection molding • Extrusion	Standard plastics textbooks & lab manuals
Sustainability	Environment	Plastics recycling • Circular economy	Research papers & sustainability curricula
Study activities	Learning needs	Exam preparation • Assignments • Lab reports	Student learning outcomes & course requirements
Programs	Education levels	BSc Polymer Engineering • MSc Plastics Engineering • PhD Materials Science	University study program catalogs
Institutions	Academia	Institute of Materials Science • Plastics research institute	University and research institute naming conventions
Users	Target groups	Students • Professors • Researchers	Academic ecosystem & stakeholder analysis
AI value	Educational role	Academic knowledge assistant • Study companion	Educational technology & AI-in-education literature



THE OFFER

A GUIDE TO **PLASTICS AI**



SPECIFIC GUIDELINE BROCHURE

 **plastics.ai**

OFFER A: CAMPUS LAB LICENSE

Aspect	Details
Offer type	Institutional infrastructure license
Product	Plastics.ai
Access scope	Unlimited use on all computers in selected labs
User type	Bachelor & Master students
Usage context	Lab courses • Project classes • Design studios
Account model	No individual login • Multi-user
Administration	Central IT deployment
Academic positioning	Official academic & industrial software

OFFER A: PRICING TABLE

Lab Size	Typical Coverage	Indicative Annual Price
Small	10–15 devices	€1,500 – €3,000
Medium	25–40 devices	€3,000 – €5,000
Large	Department labs	€5,000 – €8,000+

OFFER B: ACADEMIC ACCESS PACK

Aspect	Details
Offer type	Scalable individual access
Product	Plastics.ai
Access method	University email domain or access codes
User groups	Students • PhD candidates • Researchers
Access location	Personal devices (on/off campus)
Usage context	Exam prep • Assignments • Thesis • Research
Licensing model	Seat-based, annual

OFFER B: PRICING TABLE

Seat Tier	Typical Use Case	Indicative Annual Price
100 seats	Small department	€2,000 – €3,000
250 seats	Medium faculty	€4,000 – €6,000
500 seats	Large faculty	€7,000 – €10,000
1,000 seats	University-wide pilot	€12,000 – €16,000

OFFER C: CONCEPT OVERVIEW

Dimension	Description
Offer Name	Plastics AI & eLearning Academic Package
Core Products	Plastics.ai + Hanser plastics eLearning & eBooks
Target Users	Students • Researchers • Professors
Purpose	Support study, exam prep, labs, and research without changing teaching methods
Teaching Model	Professors keep full control of lectures, books, and curriculum
Role of AI	On-demand academic assistant aligned with trusted plastics literature

OFFER C: PACKAGE STRUCTURE

Package	Designed For	Core Focus
Essential	Departments / small cohorts	Study & exam support
Advanced	Faculties / multiple programs	Study + labs + research
Comprehensive	Strategic academic partners	Full academic support ecosystem

OFFER C: PACKAGE DETAILS

Feature Area	Essential	Advanced	Comprehensive
Plastics.ai access	✓ Department-level seats	✓ Faculty-level seats	✓ Campus-wide seats
Student usage	Study & exam prep	Study, labs, projects	Full academic usage
Researcher usage	Limited	Full research groups	Institute-wide
AI answers with references	✓	✓	✓
Priority support	—	✓	✓✓
PhD sponsorship seats	—	Optional	Included
Hanser plastics eLearning	Selected titles	Extended plastics set	Full plastics portfolio
Hanser eBooks (plastics)	Recommended reading list	Curated reference pack	Full reference access
Teaching method impact	None	None	None
Indicative annual price	€7k – €10k	€11k – €16k	€18k – €28k+

PRICING LOGIC

Pricing Component	Basis
Plastics.ai reference	€180 / user / year (public single-user price)
Institutional model	Seat-based / department / faculty licensing
Academic discount	40–70% depending on scale
eLearning & eBooks	Bundled academic licensing
Final pricing	Annual institutional quote

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CONCLUSION

CONCLUSION

- The persona analysis reveals **clear and recurring pain points** across academia: fragmented literature search, time pressure in teaching and research, complex lab preparation, exam-focused learning needs, and strict IT and data-protection requirements.
 - **Professors and researchers** struggle with fast access to **reliable, citable plastics knowledge** for lectures, supervision, and research design.
- *Plastics.ai directly addresses this with expert-validated, reference-based AI answers.*
- **Students** face difficulties in exam preparation, lab troubleshooting, and understanding complex plastics concepts.
→ *Plastics.ai provides continuous, on-demand academic support aligned with textbooks and standard curricula.*
 - **IT and university administrations** require secure, compliant, and easy-to-manage solutions.
→ *Campus and institutional licensing models meet infrastructure, privacy, and scalability expectations.*
 - The proposed **Offer A, B, and C** translate these needs into **practical, scalable solutions**, enabling campus-wide access, flexible individual usage, and long-term academic partnerships.
 - Overall, Plastics.ai successfully converts identified persona pain points into **clear academic value**, positioning itself as a **trusted AI infrastructure component** within the university ecosystem.

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THANK YOU FOR
YOUR ATTENTION!